# Andrew McKnight

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## Education

**Old Dominion University** · M.Sc., Computer Science **2013**

- Thesis-track research in bioinformatics, robotics and 3D image meshing applied to protein folding
- Three published papers with two conference presentations, winning multiple awards
- C++ programming teaching and lab assistant for three semesters

**Old Dominion University** · B.Sc., Computer Science **2012**

- Graduate-level versions of courses as part of the Accelerated Master's program
- Minor in actuarial mathematics, up to advanced probability, diff-eq and multivariate calculus

## Experience

**Sentry** · Senior Software Engineer · Remote **Nov 2021 – Current**

- *Stack*: Objective-C++, Swift, Python, SwiftUI, Django, Claude Code, SPM, XcodeGen, CocoaPods, Carthage, SwiftLint, clang-format, CodeCov, pre-commit, Fastlane, Arkana, GitHub Actions, Redash, SauceLabs, TestFlight, Linear
- *Deliverables*
  - Seer for iOS and macOS: proposed and built native apps to manage our agentic offerings
  - Sentry Cocoa SDK: app monitoring SDK for Sentry (200K orgs, 1M+ developers)
    - Integrated Specto's profiler into Sentry's Cocoa SDK and iterated with new API/features
    - Built User Feedback widget, the first user-facing UI shipped in Sentry Cocoa SDK
    - Parallelized release CI job from 35 to 5; split test suites to fix statefulness issues
    - Built demo apps for SDK and Sales teams flexible enough to simulate any customer configuration
  - Billing: platformized legacy system for product teams to self-serve how they charge for features

**Specto** · Senior Software Engineer · Remote **Nov 2019 – Nov 2021**

- *Stack*: Objective-C++, Swift, Go, Python, Protobuf, Jinja, XcodeGen, Bazel, pre-commit, GitHub Actions, Google Cloud Run
- *Highlight*: First employee at startup and remained through acquisition by Sentry
- *Deliverable*: Specto SDK, mobile app profiling SDK for iOS and Android
  - Shared C++ core between iOS and Android with Protobuf RPC to Python/Go backend
  - Hardened SDK init sequence, app launch profiling, NSURLProtocol hooks, MetricKit ingestion
  - Python scripts to generate scoped Xcode projects from Bazel targets (pre `rules-xcode`)
  - Smoke test generation validating integration across CocoaPods, Carthage, Bazel and SPM
  - Signal and mach exception propagation with open source contributions to other crash reporters

**Rightpoint** · Senior Software Engineer · Remote · Contract-to-Perm **Jan – Oct 2019**

- *Stack*: Swift, SwiftGen, SwiftLint, Sourcery, CircleCI, Postman, Browserstack, Countly
- *Deliverable*: Security app and SDK (under NDA)

**Insight Therapeutics** · Lead iOS Developer · Remote · Contract (Part-time) **Nov 2011 – Jun 2019**

- *Stack*: Objective-C, Swift, CoreGraphics, CoreData, MessageUI, UICollectionView, FMDB, SQLite, Storyboards, SSZipArchive, RNCryptor, cocoapods-keys, Trello, BitBucket, AWS
- *Highlights*
  - Sole consultation and development of multiple apps shipped in App Store and internally
  - Led client-server design with backend engineer, and UX design with clinical admins
- *Deliverables*
  - eConsent: HIPAA-compliant informed consent app for clinical trial participants
    - Internal deployment to hundreds of facilities and thousands of study subjects
    - Secure admin mode for saving/resuming progress and managing completed consents
    - Quiz-based presentation of study details with participant signature collection and export
  - Antibiotic Kinetics: antibiotic dosage calculator

*Insight Therapeutics experience continued on next page*

  - Ported from partner developer's Android app codebase, including adding first tests
  - Improved UX for error reporting through multistep calculations and data entry
- DementiaRx: public health reference e-pamphlet about dementia
  - First new app created by me shipped to the App Store for a contracting engagement
  - Scripted automatic content extraction from original paper publication PDFs
  - Built search, bookmarking and interactive patient eval elements later backported to ClotRx
- ClotRx: public health reference e-pamphlet about anticoagulants
  - Preexisting hybrid app with webviews and a PDF viewer, modernized by me
  - Migrated UIWebView→WKWebView, added search and bookmarking, redesigned for dark mode
  - Built interactive clinical patient eval modules with print/email reporting capability

**121nexus** · Software Engineer · Remote · Contract (Part-time) **Jan – Apr 2018**

- *Stack*: Swift, AVFoundation, OpenCV, Firebase, cocoapods-keys, AWS
- *Deliverable*: 121 Scan, medical inventory management app/SDK with custom barcode/QR scanning
  - Optimized critical barcode scanning component for high throughput pass/fail operation
  - Traced silent scan failures to an AVFoundation defect with double-FNC1 GS1 barcodes
  - Public launches via App Store and a white-labeled enterprise deployment in logistics facilities

**Layer** · Software Engineer · Remote **Dec 2016 – Jul 2017**

- *Stack*: C, Objective-C, Swift, OCMock, Specta, Expecta, Rails, Thrift, WebSockets, SPDY, SQLite, CocoaPods, Jenkins
- *Deliverable*: four-layer SDK offering for data→widgets→prebuilt module→sample app
  - Migrated network protocol with Erlang backend from SPDY to WebSockets under Thrift RPC
  - Rearchitected auth and transport layers as state machines using TransitionKit
  - Separated unit, integration and end-to-end tests to tighten development iteration loops
  - Streamlined developer environment configuration (Rake, Gems, CocoaPods and git submodules)

**Twitter** · Software Engineer · Cambridge, MA **Apr 2015 – Dec 2016**

- *Stack*: Objective-C, C, Swift, Java, Scala, Clojure, Coffeescript, Rails, React, Protobuf, Storm, Heron, Cascading, Scalding, Graphite, Beanstalk, Kafka, Redis, Cassandra, AWS, TeamCity, Chef, Trello, Vertica, Zeppelin
- *Highlights*
  - Presented NSOperation rearchitectures and open source tools at MobileDevCon 2016 in Amsterdam
  - Worked on high-volume lambda backend, including on-call and migration from Storm to Heron
- *Deliverables*
  - Fabric SDK (deployed to 2+ billion devices) and macOS app: developer tool ecosystem management
    - Rearchitected Apple SDK launch sequence using queues of async/compound NSOperations
    - Streamlined build and release processes with TeamCity agents deployed via Chef
  - Crashlytics SDK: crash reporting and beta deployment SDK for Apple platforms and Android
    - Researched and built feature surfacing solutions to detectable ecosystem-wide crash causes
    - Added Bitcode and tvOS compatibility to SDKs, and investigated watchOS support
    - Rearchitected CLIs to bootstrap projects, upload dSYMs and interop with our macOS app

**MC10** · Software Engineer · Cambridge, MA **Jan – Apr 2015**

- *Stack*: Objective-C, Swift, CoreBluetooth, NSOperation, TeamCity, Fabric
- *Highlights*
  - Rearchitected SDK and app codebases with NSOperation queues managing BTLE link
  - Drove adoption of Agile practices, working directly with PMs and via company-wide presentations
  - Introduced engineering practices via Git repos, GitHub and CI/CD pipelines with TeamCity/Fabric
- *Deliverable*: iOS SDK and companion app with BTLE link to proprietary medical wearable hardware

**Raizlabs** · Software Engineer · Boston, MA **Jul 2013 – Dec 2014**

- *Stack*: Objective-C, Swift, EventKit, KVO, Mustache, Core Data, QuickLook, MessageUI, WKWebView, Jenkins, TeamCity, Crittercism, Appboy, Urban Airship, CoreMetrics, Uncrustify
- *Deliverables*
  - Care.com: service seeker/provider marketplace with over 14 million members
    - iOS 7 redesign, with custom perspective effect in UIPageViewController with CATransform3D
    - Added floating quick-post button, swipeable UITableViewRowAction, UIDataDetectorTypes
    - Improved pull-to-refresh, data entry with inputAccessoryView, UIAlertView presentation/routing
  - B&H Photo: e-commerce app for a high end brick-and-mortar merchant
    - Helped rebuild B&H's iPhone app on the new iPad codebase as a universal iOS 7 app
    - Push notifications and deep linking via Appboy, crash reporting and breadcrumbs via Crittercism
    - Custom parallactic scrolling of table view headers, and barcode scanning for product search
    - Custom UICollectionView layouts and animated transitions between grid and row layouts
  - UniFirst: interactive enterprise sales catalog on iPad
    - Improved drag-and-drop clothes modeler and emblem designer built with Auto Layout and KVO
    - Generated sales proposal PDFs from Mustache HTML templates, with model/emblem snapshots
    - Added Core Data model deep copying and orphan cleanup, enabling editing existing sales orders
  - Invaluable: realtime luxury auction platform on iPad
    - Refactored critical bidding calculation routines and corrected mathematical errors
    - Added pull-to-request footer in infinite scroll UITableView, horizontal UICollectionView scroll
    - Fixed statefulness issues with UI elements dependent due to race conditions in server links

**ODU Systems Lab** · Software Engineer · Norfolk, VA · Intern **Oct 2012 – May 2013**

- *Highlight*: Supervised two international students and led consulting meetings with clients on site
- *Deliverables*
  - ChirpXM: location based social networking app based on zip code centroid radius
  - Prototyped EMR iPad app for CHKD, with basic demographic and health chart input and display

**Intergraph** · Software Engineer · Hampton, VA · Intern **May – Oct 2012**

- *Stack*: C#, VisualBasic, SQL, QF-Test, Coverity, Visual Studio, TeamFoundation
- *Deliverable*: SmartMarine/SmartPlant3D, CAD system for designing ships and power plants, with multilayer blueprint export
  - Unit and UI testing, numerical modules baseline regression testing, and montecarlo simulation
  - Manual acceptance testing for new features and fixes; database migrations with handwritten SQL

**Old Dominion University** · Project co-lead · Norfolk, VA · Capstone **Aug 2012 – May 2013**

- *Stack*: Objective-C, OpenEars, MapQuest
- *Highlights*
  - Co-lead of winning seven-member team (out of four total) and sole developer of our iOS app
  - Four presentations given to visiting professors and industry executives, including demos and Q&A

**Old Dominion University** · Research Assistant · Norfolk, VA **Aug 2012 – May 2014**

- *Stack*: Java, C++, CGAL, Boost, Jama
- *Highlights*
  - Image processing, robotics and machine learning applications in bioinformatics
  - Multiple publications and conference presentations, winning multiple awards

**Old Dominion University** · Lab Instructor · Norfolk, VA **Aug 2012 – May 2014**

- Online delivery of CS 333: Problem solving and programming in C++
- Lab instructor for CS 250: Intermediate programming in C++
- Assistant instructor for CS 150: Introduction to programming in C++
- Website maintenance, grading, office hours

**XVD Longboards** · Web Developer · Norfolk, VA · Contract **Oct 2010 – Apr 2011**

- *Stack*: ActionScript, Flash Studio
- *Highlight*: Paid with a couple custom longboards, which I got to help build myself!
- *Deliverable*: Flash website with longboard designer applying custom graphics to multiple board shapes

## Publications

- *Protein Loop Length Estimation from Medium Resolution CryoEM Images (M.Sc. thesis)* · DOI 10.25777/cxy3-9791
- *Estimating loop length from CryoEM images at medium resolutions* · DOI 10.1186/1472-6807-13-S1-S5
- *Estimating Lower Bounds on the Length of Protein Polymer Chain Segments using Robot Motion Planning*
- *CryoEM Skeleton Length Estimation using a Decimated Curve* · DOI 10.1109/BIBMW.2012.6470283

## Honors & Awards

- **2013** · **Gene Newman Award** · Virginia Modeling, Analysis and Simulation Center Capstone conference · Best overall paper
- **2012** · **Graduate with Distinction** · Old Dominion University · Awarded with B.Sc.
- **2012** · **Undergraduate Research Award** · ODU Spring Undergraduate Research Fair · Outstanding undergraduate achievement
- **2011** · **First Place** · ODU-ACM Programming Contest
- **2010 – 2011** · **Dean's List** · Old Dominion University
- **2009** · **Summa Cum Laude** · Piedmont Virginia Community College

## Presentations

**iOSDevCampCO** · Colorado Springs, CO **May 2018**

*Colloquial Swift* (45 min)

- Presented findings from analysis of common function implementations in open source repositories as a way to generate possible Swift Evolution proposals

**CocoaHeads Boston** · MIT, Cambridge, MA **Jun 2017**

*UIViewController Decomposition and UI Reuse* (20 min)

- Discussing strategies to separate responsibilities from UIViewControllers and a library of reusable UI components, e.g. animated transparent modal presentation with a close button

**Boston Algorithms Meetup** · Twitter, Cambridge, MA **Jan 2017**

*HyperLogLog* (20 min)

- Helped organized this initial meeting of the group
- Covered the theoretical workings of the HyperLogLog algorithm, with a copresenter giving a demo

**mDevCon** · Amsterdam, NL **Mar 2016**

*Command and NSOperation* (45 min)

- Delivered presentation on the Command software design pattern, and NSOperation
- Introduced subclasses we open sourced at Crashlytics for asynchronous and compound operations
- Previously delivered at a lunchtime tech talk at Twitter, Cambridge, MA (Nov 2015) and CocoaHeads Boston meetup, MIT, Cambridge, MA (Dec 2015)

**VMASC Capstone Conference** · Suffolk, VA **Apr 2013**

*Estimating lower bounds on the length of protein polymer chain segments using robot motion planning*

- Best paper award on research using motion planning algorithms to model protein molecule structures

**IEEE WCSB** · Philadelphia, PA **Oct 2012**

*CryoEM skeleton length estimation using a decimated curve*

- Using a dynamic graph algorithm to predict sequences of protein molecule substructures from 3D images

**ODU Spring Undergraduate Research Symposium** · Norfolk, VA **Apr 2012**
*CryoEM skeleton length estimation using a decimated curve*
- Developed further as a submission that was accepted to the IEEE WCSB later that fall

## Mentorship

**Girls Who Code** · Twitter, Cambridge, MA **Jul 2016**
*iOS Dev Session*
- Part of GWC summer internship covering programming first principles to full-stack style development
- Designed and led iOS session where we built a Flappy Bird clone, introducing them to Xcode as an IDE, Interface Builder, Playgrounds and Swift

**NeighborNest** · Twitter, San Francisco, CA **Oct 2015**
*Web Dev Session*
- Community program to help advocate technology literacy
- Assisted in session where students worked on a Twitter clone, teaching functional programming concepts, Javascript, and some front- and back- end concepts